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MODULAR BED FRAME ASSEMBLY

Abstract

A modular bed frame assembly is provided. The modular bed frame assembly includes a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame, each of the bed frame modules being configured to support at least a portion of a bed mattress and comprising a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between a storage configuration and an assembled configuration.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application Ser. No. 63/561,481, filed Mar. 5, 2024, and entitled MODULAR BED FRAME ASSEMBLY, the contents of which is incorporated in its entirety herein.

TECHNICAL FIELD
[0002] The technical field generally relates to furniture, and more specifically to modular bed frame assemblies.
BACKGROUND
[0003] Traditional bed frames suffer from a number of disadvantages including the difficulty of transporting the bed frame during delivery thereof and handling of bulky structural components during assembly of the bed frame. Accordingly, existing bed frames are not normally configured to be assembled by a single person and even if they are, the task can be a cumbersome one. Moreover, the bulky structural components render traditional bed frames difficult to displace to a different space.
[0004] Finally, existing bed frames typically require the use of additional hardware and tools which may make their assembly relatively complex and/or time-consuming.
[0005] In view of the above, there is a need for a bed frame which would be able to overcome or at least minimize some of the above-discussed prior art concerns.
SUMMARY
[0006] According to one aspect, there is provided a modular bed frame assembly comprising a plurality of bed frame modules, each of the bed frame modules including an inner subassembly and a peripheral subassembly configured to be secured together to form the corresponding bed frame module, wherein the inner and peripheral subassemblies are configurable between a storage configuration with panels of the corresponding one of the inner and peripheral subassemblies extending parallel to one another, and an assembly configuration wherein the panels of the corresponding one of the inner and peripheral subassemblies extend substantially normal to one another.
[0007] According to another aspect, there is provided a modular bed frame assembly comprising a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame assembly, each of the bed frame modules defining at least a portion of a mattress receiving surface and comprising a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between a storage configuration and an assembled configuration.
[0008] In certain embodiments, each of the first and second subassemblies comprises first and second panels.
[0009] In certain embodiments, when the modular bed frame assembly is assembled, each of the first and second panels of the first subassembly defines a portion of a peripheral wall of the modular bed frame assembly.
[0010] In certain embodiments, the first panel of the first subassembly comprises a slat-supporting rail.
[0011] In certain embodiments, when the modular bed frame assembly is assembled, each of the

first and second panels of the second subassembly defines a portion of a central frame portion of the modular bed frame assembly.

[0012] In certain embodiments, when the modular bed frame assembly is assembled, the first panel of the second subassembly defines a portion of a center support beam of the central frame portion.

[0013] In certain embodiments, the first panel of the second subassembly includes a slat-supporting rail.

[0014] In certain embodiments, when at least one of the first and second subassemblies is in the storage configuration, the respective first and second panels extend substantially parallel to one another.

[0015] In certain embodiments, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels extend at an angle to one another.

[0016] In certain embodiments, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels are disposed orthogonally to one another.

[0017] In certain embodiments, the first panel is pivotably secured to the second panel, at least one of the first and second subassemblies including a locking hinge configured to retain the respective first panel at a given angle relative to the respective second panel when the corresponding one of the first and second subassemblies is in the assembled configuration.

[0018] In certain embodiments, the first subassembly includes an upholstered external surface.

[0019] In certain embodiments, the modular bed frame assembly further comprises a modular headboard, the modular headboard including a plurality of headboard subassemblies, each of the plurality of headboard subassemblies being securable to a corresponding one of the plurality of bed frame modules.

[0020] In certain embodiments, the modular headboard further includes a headboard cover sized to conceal the plurality of headboard subassemblies when the modular bed frame assembly is assembled.

[0021] In certain embodiments, each of the plurality of bed frame modules is disposed sequentially with a first adjacent one of the bed frame modules along a transverse direction of the modular bed frame assembly and disposed sequentially with a second adjacent one of the plurality of bed frame modules along a longitudinal direction of the modular bed frame assembly.

[0022] According to another aspect, there is provided a modular bed frame assembly comprising a plurality of central subassemblies adapted to be assembled to form a central frame of the modular bed frame assembly, and a plurality of peripheral subassemblies, each of the peripheral subassemblies being configured to be secured to the central frame of the modular bed frame assembly to define at least a portion of a peripheral wall of the modular bed frame assembly, wherein each of the central and peripheral subassemblies is configurable between a storage configuration and an assembled configuration.

[0023] In certain embodiments, each of the central and peripheral subassemblies comprises first and second panels.

[0024] In certain embodiments, the first panel is pivotably secured to the second panel.

[0025] In certain embodiments, when the corresponding one of the central and peripheral subassemblies is in the storage configuration, the first and second panels extend substantially parallel to one another.

[0026] In certain embodiments, when the corresponding one of the central and peripheral subassemblies is in the assembled configuration, the first and second panels extend at an angle to one another.

[0027] According to another aspect, there is provided a method of assembling a modular bed frame assembly, the method comprising assembling a first bed frame module and a second bed frame module, wherein assembling the first and second bed frame modules comprises transitioning a first subassembly and a second subassembly of the corresponding first and second bed frame modules

from a storage configuration to an assembled configuration, and assembling the first and second bed frame modules in an adjacent configuration to form the modular bed frame assembly.
[0028] In certain embodiments, transitioning the first subassembly from the storage configuration to the assembled configuration comprises pivoting a first panel of the first subassembly from a position parallel to a second panel of the first subassembly, to a position at an angle to the second panel of the first subassembly.
[0029] In certain embodiments, assembling the first bed frame module comprises positioning the first and second panels of the first subassembly to define a peripheral wall of the modular bed frame assembly when the modular bed frame assembly is assembled.
[0030] In certain embodiments, assembling the first bed frame module further comprises positioning a first panel of the second subassembly and a second panel of the second subassembly to define a central frame portion of the modular bed frame assembly when the modular bed frame assembly is assembled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] FIG. **1** is a top, front perspective view of a modular bed frame assembly, in accordance with an embodiment.
[0032] FIG. **2** is a top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. **1**.
[0033] FIG. **3** is an exploded top, front perspective view of a bed frame module of the modular sofa bed assembly illustrated in FIG. **1**.
[0034] FIG. **4** is a partially exploded top, front perspective view of the modular sofa bed assembly illustrated in FIG. **1**.
[0035] FIG. **5** is a partially exploded top, front perspective view of the modular sofa bed assembly illustrated in FIG. **1**.
[0036] FIG. **6** is a partially exploded top, front perspective view of the modular bed frame assembly illustrated in FIG. **1**, shown with a first headboard module.
[0037] FIG. **7** is a partially exploded top, front perspective view of the modular bed frame assembly illustrated in FIG. **1**, shown with first and second headboard modules.
[0038] FIG. **8** is a top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. **1**, in accordance with another embodiment.
[0039] FIG. **9** is a top, front perspective view of a modular bed frame assembly, in accordance with another embodiment.
[0040] FIG. **10** is a top, front perspective view of subassemblies of the modular sofa bed assembly illustrated in FIG. **9**, the subassemblies being shown in a storage configuration.
[0041] FIG. **11** is a top, front perspective view of the subassemblies of the bed frame modules of FIG. **10**, shown in an assembled configuration.
[0042] FIG. **12** is an exploded top, front perspective view of bed frame modules of the modular sofa bed assembly illustrated in FIG. **9**.
[0043] FIG. **13** is a partially assembled top, front perspective view of the bed frame modules of the modular sofa bed assembly illustrated in FIG. **9**.
[0044] FIG. **14** is an exploded top, front perspective view of a subassembly of a bed frame module, in accordance with an embodiment.
[0045] FIG. **15** is a top, front perspective view of the subassembly of FIG. **14**.
DETAILED DESCRIPTION
[0046] It will be appreciated that, for simplicity and clarity of illustration, where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or

analogous elements or steps. In addition, numerous specific details are set forth in order to provide a thorough understanding of the exemplary embodiments described herein. However, it will be understood by those of ordinary skill in the art, that the embodiments described herein may be practiced without these specific details. In other instances, well-known methods, procedures and components have not been described in detail so as not to obscure the embodiments described herein. Furthermore, this description is not to be considered as limiting the scope of the embodiments described herein in any way but rather as merely describing the implementation of the various embodiments described herein.

[0047] For the sake of simplicity and clarity, namely so as to not unduly burden the figures with several references numbers, not all figures contain references to all the components and features, and references to some components and features may be found in only one figure, and components and features of the present disclosure which are illustrated in other figures can be easily inferred therefrom. The embodiments, geometrical configurations, materials mentioned and/or dimensions shown in the figures are optional, and are given for exemplification purposes only.

[0048] Moreover, it will be appreciated that positional descriptions such as “above”, “below”, “top”, “bottom”, “forward”, “rearward” “left”, “right” and the like should, unless otherwise indicated, be taken in the context of the figures and correspond to the position and orientation in the modular table assembly and corresponding parts when being used. Positional descriptions should not be considered limiting.

[0049] Referring to FIG. **1**, there is shown a modular bed frame assembly **10**, in accordance with one embodiment. In this embodiment, the bed frame assembly **10** is a rectangular construction defining a substantially planar mattress receiving surface **12** configured to directly support a mattress (not shown). When assembled, the modular bed frame assembly **10** is sized to receive one or more users in a supine position. For instance, the illustrated modular bed frame assembly **10** has a length of about 202 cm and a width of about 136 cm (commonly referred to as a “Queen” size bed) however, in other embodiments, the modular bed frame assembly **10** can have any other suitable dimensions.

[0050] Referring now to FIG. **2**, the modular bed frame assembly **10** includes a plurality of bed frame modules **100** that can be assembled together to form at least a portion of the modular bed frame assembly **10**. In the illustrated embodiment, the bed frame assembly **10** includes four bed frame modules **100** configured in a square configuration, i.e., with each of the bed frame modules **100** having two neighboring bed frame modules **100** positioned proximate two adjacent sides thereof. In this manner, each of the bed frame modules **100** may therefore define a peripheral corner **16** of the modular bed frame assembly **10** when the bed frame assembly **10** is assembled. In the illustrated embodiment, each of the bed frame modules **100** is disposed sequentially with an adjacent one of the bed frame modules **100** along a transverse direction X of the bed frame assembly **10** (shown in FIG. **1**) and disposed sequentially with another adjacent one of the bed frame modules **100** along a longitudinal direction Y of the bed frame assembly **10** (shown in FIG. **1**).

[0051] It is to be understood that the expression “modular” as used herein refers to a construction of the bed frame assembly **10** using interchangeable standardized components (i.e. the bed frame modules **100**) which can be assembled to form the modular bed frame assembly **10**. In accordance with some embodiments, the modular bed frame assembly **10** can therefore be assembled using a plurality of bed frame modules **100** and, if necessary or desired, reassembled to replace one or more of the bed frame modules **100** with replacement bed frame modules having a like-configuration.

[0052] In accordance with certain non-limitative embodiments, each of the bed frame modules **100** includes a central base panel **120** extending substantially horizontally and a plurality of side panels **130** secured to the central base panel **120** to define a peripheral wall of the corresponding bed frame module **100**. For instance, in the illustrated embodiment, the central base panel **120** defines a bottom portion of the bed frame module **100** with the side panels **130** extending substantially upwardly therefrom. It will be understood that, in other embodiments, the bed frame modules **100** can be arranged in any other suitable manner with, for instance, the central base panel **120** defining an upward portion of the bed frame module **100** with the side panels **130** extending substantially downwardly therefrom. In other embodiments still, the central base panel **120** can be omitted with the side panels **130** forming the structure of the corresponding bed frame module **100**.

[0053] In certain embodiments, the side panels **130** can include two peripheral side panels **132** positioned along two adjacent sides of the corresponding bed frame module **100** to define, when the modular bed frame assembly **10** is assembled, a portion of an external peripheral wall **22** of the bed frame assembly **10**. The side panels can further include two central side panels **134** positioned along the opposite two adjacent sides of the corresponding bed frame module **100**. Configured in this manner, each of the central side panels **134** can extend from an adjacent one of the external peripheral walls **22** towards a central region **18** of the modular bed frame assembly **10**, i.e., a region of the modular bed frame assembly **10** located inwardly from the peripheral wall **22** that is substantially concealed when the modular bed frame assembly **10** is assembled. In some embodiments, each of the side panels **130** can be secured to one or more adjacent ones of the side panels **130** and/or to the base panel **120** using one or more frame connectors **30**. For instance, the frame connectors **30** can include and fastener suitable for securing the side panels **130** including, for instance, brackets, clamps, screws, an adhesive, or the like.

[0054] In certain embodiments, the bed frame assembly **10** can further be configured to at least partially prevent a lateral displacement of the mattress when said mattress is positioned on the mattress receiving surface **12**. More specifically, in accordance with certain embodiments, one or more of the peripheral side panels **132** can be sized to extend upwardly above the mattress receiving surface **12** to define a peripheral edge **14** along at least a portion of a periphery of the mattress receiving surface **12**. In certain embodiments, the peripheral edge **14** can thus limit a lateral displacement of the mattress resting upon the mattress receiving surface **12**.

[0055] In some embodiments, one or more of the side panels **130** can include a slat-supporting rail **136** configured to at least partially support a slat assembly **80** (shown in FIG. **4**). In the embodiment illustrated in FIG. **3**, each of opposing ones of a peripheral side panel **132** and a central side panel **134** includes a slat-supporting rail **136** to cooperatively support a portion of the slat assembly **80**. In the illustrated embodiment, the peripheral side panel **132** and the inner side panel **134** can be disposed opposite one another and extend in the longitudinal direction Y of the bed frame assembly **10** to support opposite sides of the slat assembly **80**. It is understood that, in other embodiments, any other suitable ones of the side panels **130** can include the slat-supporting rail **136** for supporting the slat assembly **80**.

[0056] Referring to FIG. **4**, in some embodiments, adjacent ones of the bed frame modules **100** of the modular bed frame assembly **10** can be configured to cooperatively support the slat assembly **80**. For instance, adjacent bed frame modules **100** aligned in the longitudinal direction Y can cooperatively support a single slat assembly **80**. It will be appreciated that, in this manner, a user resting on the modular bed frame assembly **10** can be supported by a single slat assembly **80**. In such embodiments, the central side panel **134** extending along the transverse direction X of the modular bed frame assembly **10** can sized to provide an unobstructed extension of the slat assembly **80** across the adjacent bed frame modules **100**. More specifically, a height of the central side panel **134** extending in the transverse direction X of the modular bed frame assembly can be lower than a height of the slat-supporting rail **136**.

[0057] When the bed frame modules **100** are assembled as described above, the bed frame modules **100** can be positioned in an adjacent configuration in accordance with the desired shape and configuration of the modular bed frame assembly **10**. In order to assemble the modular bed frame assembly **10**, the bed frame modules **100** can be secured together using the frame connectors **30** described above or using any other suitable fastener. In other embodiments, one or more of the bed frame modules **100** may be sufficiently heavy or weighted down to maintain a fixed position of the bed frame modules **100** relative to their adjacent bed frame modules **100** thereby eliminating the need for the frame connectors **30**. In the illustrated embodiment, each of the bed frame modules **100** includes multiple frame connectors **30** configured to enable an assembly of the bed frame modules **100** to each other and/or to accessory components of the modular bed frame assembly **10**, such as a headboard, described in greater detail below. When used for connecting two adjacent ones of the bed frame modules **100**, the frame connectors **30** can be secured to the central side panels **134** or any other suitable element of the corresponding bed frame module **100**. As described above, the term "frame connector" as used herein, such as the frame connectors **30**, is intended to include any coupling mechanism configured to hold together adjacent ones of the bed frame modules **140** of the modular bed frame assembly **10** in a side-by-side relationship, or to secure an accessory component thereto. Individual ones of the frame connectors **30** may be referred to as hand screws, clips or hooking members.

[0058] When the bed frame assembly **10** is assembled, i.e., when adjacent ones of the bed frame modules **100** are positioned in a side-by-side relationship, a module gap or module seam **106** may be formed between the adjacent bed frame modules **100**. In particular, in certain embodiments, the module seam **106** may be visible along the peripheral wall **22** of the modular bed frame assembly **10** when assembled. Referring now to FIG. **5**, in some embodiments, the modular bed frame assembly **10** can further include a decorative covering **70** shaped and sized to overlay at least a portion of the peripheral wall **22** defined by the peripheral side panels **132**. The decorative covering **70** may provide a protection to the bed frame modules **100** or at least partially conceal the bed frame modules **100** (including, for instance, concealing a module seam **106** formed between adjacent ones of the bed frame modules **100**). In some embodiments, the decorative cover **70** can be a padded or upholstered construction comprising a textile exterior forming a hollow interior stuffed with a soft non-woven material to provide a protection to the peripheral side panels **132** of the bed frame modules **100**.

[0059] Referring now to FIGS. **6** and **7**, an embodiment of the bed frame assembly **10** is shown with a modular headboard **90** configured to be secured to a longitudinal upper end **17** of the modular bed frame assembly **10**. The modular headboard **90** can include one or more headboard subassemblies **92** configured to be secured to one or more of the bed frame modules **100** and/or to an adjacent one of the headboard subassemblies **92**. In such embodiments, the one or more bed frame modules **100** disposed proximate the longitudinal upper end **17** of the bed frame assembly **10** in the longitudinal direction Y of the bed frame assembly **10** can include one or more frame connectors **30** configured to matingly engage corresponding headboard connectors **94** of the headboard subassemblies **92**. In the illustrated embodiment, the frame connectors **30** of the bed frame modules **100** and the headboard connectors **94** of the headboard subassemblies **92** are male and female sliding connectors, respectively, configured to slidingly engage when the respective headboard subassembly **92** is translated downwardly for installation onto the corresponding bed frame module **100**.

[0060] In certain embodiments, when the modular headboard **90** includes multiple headboard subassemblies **92**, the modular headboard can further include a headboard cover **96** sized to conceal the plurality of headboard subassemblies **92** when the modular bed frame assembly **10** is assembled. More specifically, the headboard cover **96** can be configured to at least partially conceal the headboard subassemblies **92** (including, for instance, concealing a seam formed between adjacent ones of the headboard subassemblies) while maintaining access to the headboard connectors **94**. In some embodiments, the headboard cover **96** can be a padded or upholstered construction comprising a textile exterior forming a hollow interior stuffed with a soft non-woven material to provide a protection to the headboard subassemblies **92** of the headboard module **90**.

[0061] In the illustrated embodiment, each of the bed frame modules **100** disposed proximate the longitudinal upper end **17** of the bed frame assembly **10** is configured to be assembled with a corresponding headboard subassembly **92**. Alternatively, in other embodiments, the modular bed frame assembly **10** can include a single headboard module **90** sized to be coupled with multiple bed frame modules **100**.

[0062] In some embodiments, one or more of the side panels **130** of the bed frame modules **100** can be configured to contact a ground surface to provide a support to the corresponding bed frame module **100**. For instance, in the illustrated embodiment, each of the side panels **130** is dimensioned to abut the ground surface so as to at least partially support a weight of the bed frame module **100** as well as the mattress resting on the bed frame assembly **10** and any user(s) resting thereupon. Referring now to FIG. **8**, in other embodiments, one or more of the bed frame modules **100** can include a support member **140** configured to be secured to one or more of the base panel **120** and the side panels **130**. The support member **140** can extend downwardly therefrom to the ground surface to at least partially support the corresponding bed frame module **100**. In such embodiments, each support member **140** can include a connection portion connectable to the base panel **120** and a support portion extending away from the connection portion for contacting the ground surface.

[0063] Referring now to FIGS. **9** to **15**, there is shown a modular bed frame assembly **20**, in accordance with another embodiment. The modular bed frame assembly **20** includes a plurality of bed frame modules **200** adapted to be assembled in an adjacent configuration to form the modular bed frame assembly **20**. Notably, in contrast with the modular bed frame assembly **10** describe above, one or more of the bed frame modules **200** comprises collapsable subassemblies which can be configured in one of a storage configuration and an assembled configuration. It will be understood that, as used herein, the expression “subassemblies” is intended to refer to pre-constructed components that are assembled together to form a part of the bed frame module **200** and to provide structural strength and stability thereto. In this manner, the expression “subassemblies” is not intended to refer to smaller individual elements including fasteners used to connect or secure the subassemblies.

[0064] Referring to FIGS. **10** to **15**, in the illustrated embodiment, each of the bed frame modules **200** includes a first subassembly **210** and a second subassembly **220** configured to be assembled to form the corresponding bed frame module **200**. In the illustrated embodiment, the first subassembly **210** includes first and second panels **212**, **214** which, when the modular bed frame assembly **20** is assembled, each define a portion of a peripheral wall **22** of the modular bed frame assembly **20**. For instance, in the embodiment illustrated in FIG. **13**, the first panel **212** of the first subassembly **210** defines a portion of a side rail **23** of the modular bed frame assembly **20** (shown in FIG. **9**) extending along a longitudinal axis Y thereof. In certain embodiments, the second panel **214** of a given one of the bed frame modules **200** disposed at a longitudinal lower end **25** of the modular bed frame assembly **20** can define a portion of a footboard **29** of the modular bed frame assembly **20** extending along a transverse axis X thereof. In such embodiments, the first subassembly **210** can thus be referred to as a “peripheral subassembly” and the first and second panels **212**, **214** can be referred to as “peripheral panels”.

[0065] In certain embodiments, when the first subassembly **210** is a peripheral subassembly, the second subassembly **220** can include first and second panels **222**, **224** defining a portion of a central frame **26** of the modular bed frame assembly **20**. In such embodiments, the second subassembly **220** can be referred to as a central subassembly and the first and second panels **222**, **224** can be referred to as central panels. In certain embodiments, the central frame **26** of the modular bed frame assembly **20** can form an internal structural component of the modular bed frame assembly **20** configured to provide structural support and stability thereto, ensuring that all the components of the modular bed frame assembly **20** are securely held in place. In this manner, the central frame **26** can act as the foundational structure of the modular bed frame assembly **20** to distribute weight and absorb stress. The expression “internal structural component” is intended to refer to a component of the modular bed frame assembly **20** that is substantially concealed when the modular bed frame assembly **20** is assembled.

[0066] In certain embodiments, the central frame **26** of the modular bed frame assembly **20** can be self-supporting, i.e., the central frame **26** can be configured to contact the ground surface and support a weight of the modular bed frame assembly **20** when assembled. To that end, one or more of the first and second panels **222**, **224** can be sized to contact the ground surface beneath the central frame **26** and/or the central frame **26** can include one or more supporting members configured to at least partially support the central frame **26**, similar to the support members **140** described above.

[0067] In certain embodiments, the first subassembly **210** can include a pivoting connector **32** to pivotably connect the first and second panels **212**, **214**. When configured in this manner, the first and second panels **212**, **214** of the first subassembly **210** can be pivotally hinged to one another to enable the first subassembly **210** to transition between the storage configuration and the assembled configuration. The pivoting connector **32** can include a hinged connector or any other connector suitable for enabling a pivoting connection between the first and second panels **212**, **214**. Similarly, in certain embodiments, the first and second panels **222**, **224** of the second subassembly **220** can be pivotally hinged to one another via a hinged connector **32** to enable the second subassembly **220** to transition between the storage configuration and the assembled configuration.

[0068] Referring now to FIG. **10**, the second subassemblies **220** of multiple bed frame modules **200** are shown in a collapsed configuration. When in the collapsed configuration, the first and second panels **222**, **224** are pivoted such that the first panel **222** extends substantially parallel to the second panel **224**. Referring now to FIG. **11**, the second subassemblies **220** of multiple bed frame modules **200** are shown in an assembled configuration. When in the assembled configuration, the respective first and second panels **222**, **224** extend at an angle to one another. For instance, in the illustrated embodiment, the first and second panels **224**, **224** are disposed orthogonally to one another when the corresponding second subassembly **220** is in the assembled configuration.

[0069] In certain embodiments, the first subassembly **210** can be configured similarly to the second subassembly **220**. More specifically, the first and second panels **212**, **214** can be pivoted such that the first panel **212** extends substantially parallel to the second panel **214** when the first subassembly is in the storage configuration. Moreover, the first and second panels **212**, **214** of the first subassembly **210** can extend at an angle to one another when the first subassembly **210** is in the assembled configuration with, in certain embodiments, the first and second panels **212**, **214** of the first subassembly **210** extending orthogonally to one another when the first subassembly **210** is in the assembled configuration.

[0070] While the illustrated embodiment includes first and second subassemblies **210**, **220** including a pivoting connector **32**, it will be understood that, in other embodiments, the first and second subassemblies **210**, **220** can be assembled using any other type of connector including, for instance, connectors which restrict a pivoting between the corresponding first and second panels. For instance, in certain embodiments, the corresponding first and second panels can be detached to adopt the storage configuration and attached to adopt the assembled configuration.

[0071] It will be appreciated that, when the bed frame modules **200** of the modular bed frame assembly **20** are disassembled and the corresponding first and second subassemblies **210**, **220** are in the storage configuration, the modular bed frame assembly **20** may have a reduced volume with reduced dimensions in comparison to a traditional bed frame thus promoting efficient use of space and organization. Accordingly, the configuration of the bed frame modules **200** may facilitate transportation of the modular bed frame assembly **20** and the use of components having a reduced size may facilitate installation thereof.

[0072] In certain embodiments, each of the first and second subassemblies **210**, **200** includes a pivot stopper **219** configured to limit a pivoting rotation of the corresponding first and second panels beyond a predetermined angle. For instance, in some embodiments, the pivot stopper **219** can be configured to limit a pivoting rotation of the corresponding first and second panels beyond an angle of 90°. The pivot stopper **219** can be configured as a wall member, a flexible tension member or any other component configured to limit a pivoting rotation between to hinged members. For instance, in the illustrated embodiment, the pivot stopper **219** is integrated into the pivoting connector **32** provided as a locking hinge. In such embodiments, the pivoting connector **32** including the pivot stopper **219** is configured to retain the respective first panel at a given angle relative to the respective second panel when the corresponding one of the first and second subassemblies **210**, **220** is in the assembled configuration

[0073] Referring now to FIGS. **12** and **13**, as specified above, the second subassemblies **220** of the bed frame modules **200** can be secured together to form the central frame **26** of the modular bed frame assembly **20**. More specifically, the first panel **222** of each of the second subassemblies **220** can define a portion of a central support beam **27** of the modular bed frame assembly **20**. The central support beam **27** can extend longitudinally from a longitudinal upper end **24** of the modular bed frame assembly **20** to the longitudinal lower end **25** thereof. Similarly, in certain embodiments, the second panel **224** of each of the second subassemblies **220** can define a portion of a transverse support beam **28** of the modular bed frame assembly **20**. The transverse support beam **28** can extend transversely between opposing ones of the side rails **23** of the modular bed frame assembly **20**.

[0074] It will be understood that, in certain embodiments, the central frame **26** of the modular bed frame assembly **20** can help prevent mattress sagging, provide additional stability and durability, ensure an even weight distribution, and/or maintain an overall structural integrity of the modular bed frame assembly **20**.

[0075] The second subassemblies **220** can be secured together using one or more frame connectors **240** or any other suitable connector. Once the rigid inner frame **220** is assembled, one or more of the peripheral assemblies **216** can be secured thereto to form the bed frame assembly **20** (as shown in FIG. **13**).

[0076] In certain embodiments, one or more of the bed frame modules **200** can include a slat-supporting rail **230** configured to at least partially support the slat assembly **80**. In the embodiment illustrated in FIGS. **12** and **13**, each of the first panels **212**, **214** of the first and second subassemblies **210**, **220** includes a slat-supporting rail **230** to cooperatively support a portion of the slat assembly **80** when the bed frame assembly **20** is assembled. In certain embodiments, each of the bed frame modules **200** can be configured to receive a distinct slat assembly **80**. In the illustrated embodiment, the transverse support beam **28** of the central frame **26** is sized to enable an extension of the slat assembly **80** across multiple bed frame modules **200** aligned in the longitudinal direction Y of the modular bed frame assembly **20**. More specifically, a height of the second panel **224** of the second subassemblies **220** can be limited to not extend above a height of the slat-supporting rail **230**. In this manner, the slat assembly **80** can extend across multiple bed frame modules **200** aligned in the longitudinal direction Y without obstruction. It will be understood that the slat assembly **80** extending across multiple bed frame modules **200** can provide better support for the mattress by more evenly distributing a weight of the mattress across the modular bed frame assembly **20**, thus potentially further preventing sagging. The slat assembly **80** extending across multiple bed frame modules **200** can further improve airflow and reduce pressure points, contributing to a more comfortable and durable sleeping surface.

[0077] A method of assembling the modular bed frame assembly **20** will now be described in further detail. The method can include assembling multiple bed frame modules **200**. The assembly of the bed frame modules **200** can include transitioning the first subassembly **210** and the second subassembly **220** of the respective bed frame modules **200** from a storage configuration to an assembled configuration and securing the first and second subassemblies **210**, **220** together to form the corresponding bed frame module **200**. The method can further include assembling the bed frame modules **200** in an adjacent configuration to form the modular bed frame assembly **20**. In certain embodiments, the method can include securing the bed frame modules **200** together using

one or more frame fasteners.
[0078] In certain embodiments, transitioning the first and second subassemblies **210**, **220** from the storage configuration to the assembled configuration can include pivoting the first panel **212**, **222** of the first and second subassemblies **210**, **220** from a position parallel to the second panel **214**, **224** of the first and second subassemblies **210**, **220**, to a position at an angle to the second panel **214**, **224**.
[0079] In other embodiments, a method of assembling the modular bed frame assembly **20** can include transitioning the second subassemblies **220** of multiple bed frame modules **200** from a storage configuration to an assembled configuration as described above. The method can further include securing the second subassemblies **220** together to form the central frame **26** of the modular bed frame assembly **20**. In certain embodiments, the method can further include transitioning the first subassemblies **210** of the multiple bed frame modules **200** from the storage configuration to the assembled configuration as described above and securing the first assemblies **210** to the central frame **26** to define the peripheral wall **22** of the modular bed frame assembly **20**.
[0080] In certain embodiments, each of the bed frame modules **100**, **200** can define a storage compartment when assembled delimited by the corresponding side panels. In some embodiments, the storage compartment can define a storage space shaped and dimensioned to store personal belongings such as sleeping accessories including, for example, pillows, sheets and covers.
[0081] While the above description provides examples of the embodiments, it will be appreciated that some features and/or functions of the described embodiments are susceptible to modification without departing from the spirit and principles of operation of the described embodiments. Accordingly, what has been described above has been intended to be illustrative and non-limiting and it will be understood by persons skilled in the art that other variants and modifications may be made without departing from the scope of the invention as defined in the claims appended hereto.

Claims

1. A modular bed frame assembly comprising: a plurality of bed frame modules adapted to be assembled in an adjacent configuration to form the modular bed frame assembly, each of the bed frame modules defining at least a portion of a mattress receiving surface and comprising: a first subassembly and a second subassembly configured to be assembled to form the respective bed frame module, each of the first and second subassemblies being configurable between an assembled configuration suitable for the assembly of the respective bed frame module, and a storage configuration.
2. The modular bed frame assembly of claim 1, wherein each of the first and second subassemblies comprises first and second panels, the first and second panels being adjustable between the storage and assembled configurations.
3. The modular bed frame assembly of claim 2, wherein, when the modular bed frame assembly is assembled, each of the first and second panels of the first subassembly defines a portion of a peripheral wall of the modular bed frame assembly.
4. The modular bed frame assembly of claim 3, wherein the first panel of the first subassembly comprises a slat-supporting rail.
5. The modular bed frame assembly of claim 2, wherein, when the modular bed frame assembly is assembled, each of the first and second panels of the second subassembly defines a portion of a central frame of the modular bed frame assembly.
6. The modular bed frame assembly of claim 5, wherein, when the modular bed frame assembly is assembled, the first panel of the second subassembly defines a portion of a central support beam of the central frame.
7. The modular bed frame of claim 6, wherein the first panel of the second subassembly includes a slat-supporting rail.

8. The modular bed frame assembly of claim 2, wherein, when at least one of the first and second subassemblies is in the storage configuration, the respective first and second panels extend substantially parallel to one another.
9. The modular bed frame assembly of claim 8, wherein, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels extend at an angle to one another.
10. The modular bed frame assembly of claim 8, wherein, when at least one of the first and second subassemblies is in the assembled configuration, the respective first and second panels are disposed orthogonally to one another.
11. The modular bed frame assembly of claim 2, wherein the first panel is pivotably secured to the second panel, at least one of the first and second subassemblies including a locking hinge configured to retain the respective first panel at a given angle relative to the corresponding second panel when the corresponding one of the first and second subassemblies is in the assembled configuration.
12. The modular bed frame assembly of claim 1, wherein the first subassembly includes an upholstered external surface.
13. The modular bed frame assembly of claim 1, further comprising a modular headboard, the modular headboard including a plurality of headboard subassemblies, each of the plurality of headboard subassemblies being securable to a corresponding one of the plurality of bed frame modules.
14. The modular bed frame assembly of claim 13, wherein the modular headboard further includes a headboard cover sized to conceal the plurality of headboard subassemblies when the modular bed frame assembly is assembled.
15. The modular bed frame assembly of claim 1, wherein each of the plurality of bed frame modules is disposed sequentially with a first adjacent one of the bed frame modules along a transverse direction of the modular bed frame assembly and disposed sequentially with a second adjacent one of the plurality of bed frame modules along a longitudinal direction of the modular bed frame assembly.
16. A modular bed frame assembly comprising: a plurality of central subassemblies adapted to be assembled to form a central frame of the modular bed frame assembly; and a plurality of peripheral subassemblies, each of the peripheral subassemblies being configured to be secured to the central frame of the modular bed frame assembly to define at least a portion of a peripheral wall of the modular bed frame assembly; wherein each of the central and peripheral subassemblies is configurable between an assembled configuration suitable for the assembly of the modular bed frame assembly, and a storage configuration.
17. The modular bed frame assembly of claim 16, wherein each of the central and peripheral subassemblies comprises first and second panels, the first and second panels being adjustable between the storage and assembled configurations.
18. The modular bed frame assembly of claim 17, wherein the first panel is pivotably secured to the second panel.
19. The modular bed frame assembly of claim 18, wherein when the corresponding one of the central and peripheral subassemblies is in the storage configuration, the first and second panels extend substantially parallel to one another.
20. The modular bed frame assembly of claim 19, wherein when the corresponding one of the central and peripheral subassemblies is in the assembled configuration, the first and second panels extend at an angle to one another.